DANIEL SCHEUCHENSTUHL

Curriculum Vitae

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INFO

I am a Research Scientist at Siemens AG Austria with 3 years of post-MSc professional experience and a 14-year technical trajectory in software engineering augmented with deep expertise in Machine Learning, Computer Vision, Robotics, Generative AI, and MLOps. I currently specialize in Physical AI: the development of autonomous systems that perceive, reason, and act in complex, real-world environments. My work lives at the intersection of Machine Perception, Distributed Systems, and Embedded AI. As a polyglot engineer, I operate across the full robotics stack where I have strong experience in designing robust model lifecycles for latency-critical applications.

PROFESSIONAL EXPERIENCE

Research Scientist

Siemens AG Austria

October 2023 - ongoing Vienna, Austria

- Assessed the maturity and industrial readiness of SOTA Vision-Language-Action (VLA) models ($\pi_{0.5}$) across two deployment tracks: a real-world pick-and-place setup on a UR7e cobot with cross-embodiment teleoperation and a custom data pipeline, and a simulation-based counterpart using NVIDIA Isaac Sim with $\pi_{0.5}$ deployed on a Siemens AI Inference Server on an Industrial Edge Device — designing the full system architecture, data engineering stack, and closed-loop inference pipeline from scratch.
- Led the production deployment of Industrial Vision Foundation Models, achieving a 50× reduction in inference latency and accelerated data engineering workflows by 100× using Ray, enabling few-shot visual quality inspection for industrial applications.
- Architected a robust and scalable end-to-end system for edge AI, establishing engineering best practices and mentoring juniors.
- Collaborated cross-functionally with research, engineering, and product teams.

Working student

Siemens AG Austria

August 2020 - September 2023 Vienna, Austria

- Developed AI and robotics solutions to automate visual quality inspection tasks

Project assistant

Vienna University of Technology

September 2020 - July 2022 Vienna, Austria

- Contributed to research projects and held student seminars

CORE COMPETENCIES

- Rapid Prototyping
- Research-to-Production
- Scalable AI Architecture
- Cross-Functional Engineering
- Generative & Multimodal AI

EXTRACURRICULAR ACTIVITIES

F1Tenth - Autonomous driving competition

ICRA 2022 (Team Scuderia Segfault)

May 2022 Philadelphia, USA

- Third place in the time-trials and the head-to-head competition

F1Tenth - Autonomous driving competition

IROS 2021 (Team Scuderia Segfault)

September 2021 Prague, Czech Republic

- First place in the time-trials and the head-to-head competition

EDUCATION

Diplom-Ingenieur (Dipl.-Ing./MSc)

Vienna University of Technology

November 2020 - September 2023

Master program: Computer Engineering

Bachelor of Science (BSc)

Vienna University of Technology

October 2017 - November 2020

Bachelor program: Computer Engineering

Matura (A-Levels)

HTL Wels für Informationstechnologie

September 2011 - June 2016

ENGINEERING SKILLS

Key Areas

- Artificial Intelligence focusing on Embodied AI / Robotics and Machine Perception / Computer Vision (industrial applications and autonomous driving)
- Software Architecture focusing on Distributed Systems and Edge Computing

Software Engineering

- Programming Languages: Python, C, Java, C++, Bash, Go
- Frameworks/Libraries: PyTorch, Jax, TensorFlow, Isaac Sim, ROS, OpenCV, Numpy, Slurm, Ray
- DevOps/MLOps: Git, CI/CD, Docker
- Experience with distributed systems, cloud computing (distributed training) & edge computing

AI & ML

- Vision-Language-Action (VLA) models, Generative AI, Computer Vision, Foundation Models (e.g. SAM)
- Learning paradigms: Unsupervised Learning, Supervised Learning, Reinforcement Learning, Imitation Learning, Self-Supervised Learning

Automation, Embedded Systems & Hardware Engineering

- PLC & HMI programming
- Microcontrollers
- Digital Design & Computer Architecture (FPGAs)

Complementary Skills

- Formal Methods
- Extensive hands-on experience with Unix/Linux systems
- Network engineering experience with Cisco and Siemens devices

Additional Skills

- Driving license: B

HOBBIES & INTERESTS

- Running, gym, hiking, skiing, obstacle course racing (e.g. Spartan Race), gaming
- Science & technology, traveling, music, cooking, culture

LANGUAGES

German ●●●●●

English ●●●●○

Spanish ●○○○○

SELECTED PUBLICATIONS & RESEARCH

Synthetic training data generation for controllable learning of image recognition in recycling

Published @Springer Nature Link, 2026

- Developed a novel procedural alpha-matting based dataset generation approach that creates new compositions and labels simultaneously using limited real data.

Attentional Neural Network based Dynamic Object Detection for Autonomous Multi-Agent Systems

Master Thesis @TU Wien, 2023

- Developed a human-attention inspired perception pipeline for autonomous racing, demonstrating superior robustness on Out-of-Distribution (OOD) data compared to SOTA baselines.

Enhancing Robot Learning through Learned Human-Attention Feature Maps

Published @RAP4Robots workshop at ICRA, 2023

- Proposed novel feature map enhancement techniques to improve the data efficiency of robot learning agents.

Timing Analysis of TSN-Enabled OPC UA PubSub

Published @18th IEEE International Workshop on Factory Communication Systems (WFCS), 2022

- Analyzed Worst-Case Execution Time (WCET) for real-time industrial networks, ensuring deterministic performance for distributed control systems.